MILITARY SPECIFICATION

SLING SYSTEM, HELICOPTER: EXTERNAL CARGO, 40,000 POUND CAPACITY

This specification is approved for use by U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of adjustable cargo sling system used for lifting equipment and cargo by helicopter. The sling system shall be capable of supporting external helicopter loads weighing up to 40,000 pounds.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

*2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation. form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited)
PPP-B-601	- Boxes, Wood, Cleated Plywood
PPP-B-636	- Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-I-6866	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commandant of the Marine Corps, Headquarters, U.S. Marine Corps (Code LMA-1), Washington, D.C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-F-7190 - Forging, Steel, for Aircraft and Special Ordnance Applications
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attribute
Marking for Shipment and Storage
MIL-STD-810 Environmental Test Methods and Engineering Guidelines

*2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publication form a part of this specification to the extent specified herein.

DRAWINGS

U.S. Marine Corps

825029A0000 Sling, Multileg, Helicopter, External Lift, 40,000-Pound Capacity
825029A1000 Apex Shackle Subassembly
825029A2000 Sling Leg
825029A3000 Grab Link
825029A4000 Chain Leg
825029A5000 Coupling Link

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM A304 Alloy Steel Bars subject to end-quench hardenability requirements

ASTM A322 Steel Bars, Alloy, Standard Grades

ASTM D395.1-82 Packaging, Commercial

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this. specification shall take precedence.

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3.1 First article. The components from two complete sling systems as described in 3.3 shall be submitted for first article examination and tests. Unless otherwise specified in the contract, first articles shall be manufactured using the same methods and under the same conditions and procedures intended for use on production units. The first articles shall be tested as specified in Section 4 to determine that, prior to starting production, the contractor's production methods are capable of yielding sling system which meet the requirements specified herein.

3.2 Materials. Except as specified herein and on the applicable drawings and specifications, materials shall be-selected by the manufacturer. Materials selected shall be capable of meeting the performance requirements specified herein.

*3.3 Design and construction. The sling system shall consist of one apex shackle subassembly, four sling legs, four grab links, four coupling links, and four chain legs manufactured in accordance with Drawing No. 825029A0000 and this specification. The sling system shall be fabricated, assembled, and stowed in accordance with the drawings and specifications specified herein. No deviation from the prescribed dimensions or tolerances is permissible. The sling system shall conform to Drawing No. 825029A0000. Seventeen components shall make up a complete sling system; one apex shackle, four Kevlar Aramid Fiber, or equal chain legs, four coupling links, four grab links and four chain legs.

*3.3.1 Apex shackle subassembly. The apex shackle subassembly shall act as an interface between the Kevlar Aramid Fiber, or equal, sling legs and the helicopter hook,

*3.3.2 Sling legs, Each sling leg shall be 3-strand, made of Kevlar Aramid fiber, or equal, and shall be of a “wire-rope” construction as shown in Drawing No. 825029A2000. Each of the three strands shall have a braided dacron, or equal, jacket and the rope shall be covered with polyolefin, or equal, shrink tubing. The eye splices, as described in Drawing No. 825029A2000, shall be covered with shrink tubing and dacron, or equal, sock, then coated with a waterbased adhesive emulsion.

*3.3.3 Grab links. The grab links shall be connected to the Kevlar Aramid Fiber, or equal, sling legs. The coupling links specified in Drawing No. 825029A5000 shall be connected to the grab links which shall be connected to the chain legs specified in Drawing No. 825029A4000. The free ends of the chain legs shall be passed through or around the load pickup points and secured by the grab links. The length of each chain leg shall be adjustable to the load’s requirement. The working length of the chain leg depends on which. chain link is secured by the grab link.

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*3.3.4 Coupling link. A coupling link shall serve as the interface between each grab link and chain leg. the coupling links shall be as specified in Drawing No. 825029A5000.

*3.3.5 Chain leg. The chain leg specified in Drawing No. 825029A4000 shall serve as the interface between the load to be lifted and the coupling link specified in 3.3.4.

*3.4 Performance. The sling system described-herein shall provide capability to suspend and transport cargo internally beneath cargo helicopters. The sling systems are used with single, dual point, or rnultipoint suspension helicopters. Various numbers of lifting legs may be configured to converge to single hooks or be separated in various ways between fore, aft and center helicopter hooks.

3.4.1 Load capacity. The sling system (four legs) shall have a safe working load of 40,000 pounds. Reducing the number of legs in the rigged configuration reduces the maximum load capability of the sling system in direct proportion to loss of legs. However, when only leg is used in suspending the total load, the maximum allowable single leg load may be increased by a factor of 1.4 to 14,000 pounds. When rigged, the sling legs shall not exceed 45 dearees from a verticle center line projected through the helicopter hook.

*3.4.2 Load strength. The sling system components shall sustain up to the following loads in compliance with first article test and production sampling requirements and shall conform with the quality assurance requirements specified in Section 4.

Sling System Component Assemblies	Reference Drawing	Safe Working Load 1/ (lbs)	Proof Load 2/ (lbs)	Ultimate Breaking Strength Lead 3/ (lbs)
Apex Shackle Subassembly	825029A1000	40,000	100,700	151,200
Sling Leg	825029A2000	10,000	39,800	59,800
Grab Link	825029A3000	10,000	39,800	59,800
Chain Leg	825029A4000	5,000	19,900	29,900
Coupling Link	825029A5000	5,000	19,900	29,900

1/ Safe working load. Maximum load weight that component is rated to lift.

2/ Proof load. Maximum working tensile load of component during standard lifting operations.

3/ Ultimate breaking strength load. Maximum working tensile load multiplied by a safety factor of 1.5.

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*3.4.3 Setting of the shrink tubing. Heat shall be applied to shrink tubing of each sling leg while under a 23,700-pound load.

3.4.4 Environmental. Unless otherwise specified herein, the sling system shall meet the performance requirements specified herein after exposure to environmental conditions as follows:

3.4.4.1 Temperature.

3.4.4.1.1 Storage. The sling system shall withstand storage temperatures from -48°C (-55°F) to +74°C (+165°F) and thereafter meet the requirements specified in 3.4.

*3.4.4.1.2 Operating. The sling system shall be capable of meeting the requirements specified in 3.4 at any temperature from 32°C (-25°F) to 52°C (+125°F). This requirement will not be tested but is provided to assist in material selection.

3.4.4.2 Humidity. The-sling system shall withstand relative humidity up to 95 percent as specified in MIL-STD-810, method 507, procedure 1 and thereafter shall meet the requirements specified in 3.4.

*3.5 Weight. The weight of a complete sling system, one apex shackle subassembly, 4 sling legs, 4 coupling links, 4 grab links, and 4 chain legs, shall not exceed 170 pounds.

3.6 Marking. Major components described in 3.3 shall be marked for identification as shown on applicable drawings.

3.7 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of the specification and applicable drawings. All parts used in the sling system shall be free of burrs, sharp edges, chips and foreign matter. The standards of workmanship exhibited in any approved first article, subject to any qualifications stated in the Government's notice of approval, shall be demonstrative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PREVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right

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to perform any of the inspections set forth in the specification. where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article (see 4.3)

b. Production (see 4.4)

c. Quality Conformance (see 4.5)

d. Preparation for delivery (see 4.8)

4.3 First article examination and test. The components from two complete sling systems manufactured to the requirements of Drawing No. 825029A0000, shall be subjected to the examination and tests of 4.6. and 4.7. Failure of any component to pass any of the tests shall be cause for rejection of the sling system. Further production of sling systems by the manufacturer, prior to approval of the first article, shall be at the manufacturer's risk.

*4.4 Production inspection samples. From each succeeding inspection lot of 50 or less sling systems which have passed the examination of 4.6, a random sample shall be selected in accordance with MIL-STD-105, acceptable quality level (AQL) 6.5, level II, and the components shall be subjected to the applicable tests specified in 4.7. Failure of any component to pass the tests specified in 4.7 shall cause acceptance of the sling system, and the inspection lot from which the system was selected, to be withheld untilthe following have been completed:

a. Determine the cause of the failure.

b. Take the necessary action, with the concurrence of the procuring activity to correct or eliminate the failure from all units in the inspection lot and, if so directed by the procuring activity, from those units previously shipped.

c. Submit the proposed corrective action to the procuring activity for approval to prevent recurrence of the failure.

4.4.1 Inspection lot. An inspection lot is defined as a collection of 25 or less sling systems which are alike in construction and place of manufacture and have passed examination and tests specified in 4.5.

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Table I. Examination

Test	Defect	Classification Major	Minor
Materials	Any part not fabricated of specified material	x	
Design	Any characteristic not in accordance with specified requirements	x	
Cleanliness	Dirt, grease, oil, or foreign matter	x	
Workmanship	Burrs, sharp edges, surface roughness, loose construction broken rovings, frayed rope assembly, loose or torn shrink tubing, ends of eye sock not serviced.	x	
Assembly	Part missing, not assembled or joined as specified. Operation omitted or not performed as specified.	x	
Component dimensions	Perpendicularity, parallelism, concentricity, out of specified tolerance.	x	
	Out of “Unless otherwise specified” tolerance.		x
Cadmium plating	Evidence of hydrogen embrittlement, unable to document specified baking after plating per QQ-P-416	X	
Anodizing	Not as specified in MIL-A-8625	x	
Keeper spring torque	Out of tolerance at specified angular positions	x	
Identification	Not as specified on component		x

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4.5 Quality conformance examination. Each sling system shall be subjected to the visual examination in 4.6 and the tests of 4.7.3. Inability of the sling system or component to pass the examination of 4.6 and tests of 4.7.3 shall constitute a failure.

4.6 Examination. Each sling system shall be examined for conformance to the material and workmanship requirements specified in Table I.

4.7 Tests.

*4.7.1 Test conditions. Unless otherwise specified herein, the sling system shall be subjected to inspection and test under the following conditions:

a.	Temperature	Room ambient +27+8°C (80+15°F).
b.	Relative Humidity	Room ambient up to 95 percent.
c.	Altitude	Normal ground atmospheric pressure of 760 +50mm (30 + 2 inches) of mercury.

4.7.2 First article and production sample inspections. The first article and production sample sling system components shall be subjetted to the following tests after satisfactorily passing the tests and examination specified in 4.6 and 4.7.3.

4.7.2.1 Test procedure for sling system components. Subject the sling system components to the tests listed in Table II. Unless otherwise specified in the individual test paragraph, the tests required in Table II shall be performed under test conditions specified in 4.7.1. Failure of any component to pass any of the tests shall constitute a failure.

*4.7.2.2 Cyclic load (fatique) test. Sling legs shall be cyclic load (fatique) tested and loads shall be within 3 percent of the following:

Step No.	Max Load	Min Load	Hz	No. of Cycles
1	27,000 lbS	25,000 lbs	5	57,000
2	30,000 lbs	27,000 lbs	2	155
3	38,000 lbs	27,000 lbs	1	5
4	Repeat steps 1, 2, and 3 for 40 repetitions.			

*4.7.2.3 Proof load tests. The component assemblies shall sustain not less than the proof loads cited in 3.4.2. All loading shall be maintained for not less than 5 minutes after reaching the specified load. Inability to hold the specified load for at least 5 minutes, or failure or breakage of any component, shall constitute failure of this test.

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Table II. Tests

Test	Para	1st Article (4.3)	Quality Conformance (4.5)	Production Sampling (4.4)
Nondestructive examination	4.7.3	x	x	x
Cyclic load tests	4.7.2.2	x		
Proof load tests	4.7.2.3	x		x
Ultimate breaking strength load tests	4.7.2.4	x		x
Elongation tests	4.7.2.5	x		x
Net length tests	4.7.2.6	x		x

*4.7.2.4 Ultimate breaking strength load tests. The component assemblies shall sustain not less than the ultimate loads cited in 3.3 and 3.4. Inability to meet the specified load shall constitute failure of this test.

4.7.2.5 Elongation test. The sling legs assemblies shall not be elongated in excess of 7.0 inches at completion of 5 minutes holding time during the proof load test of 4.7.2.3. Elongation in excess of 7.0 inches shall constitute a failure.

*4.7.2.6 Net length test. The length of the sling leg assemblies measured between the load bearing surfaces while sling leg assembly is under 100 ± 10 pounds of tension shall be 144 inches +1.5 inches after the proof load test of 4.7.2.3.

4.7.3 Quality conformance tests. Each sling system shall have components tested in accordance with the following. Records of tests shall be submitted with each sling system.

4.7.3.1 Nondestructive examination. Inability of any component to pass the acceptance criteria (Table I) shall constitute failure of this test.

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*4.7.3.2 Magnetic particle inspection. Each grab link shall be magnetic particle inspected prior to cadmium plating using methods of MIL-I-6868.

*4.7.3.3 Flourescent inspection. Each grab link spacer shall be flourescence penetrant inspected prior to anodizing, using methods of MIL-I-6866.

*4.7.3.4 Hardness testing. Each grab link, apex shackle sub-assembly, steel spacer and steel pin shall be machined as specified herein, heat treated arid then tested for hardness. Hardness shall be Rc-38-42.

*4.7.3.5 Forging integrity. Each grab link and apex shackle sub-assembly must satisfy the forging requirements of MIL-F-7190 and forging notes of Drawing No. 825029A1000 and 825029A3000. Accept-ability shall be documented by certification of material composition and strength.

4.8 Preparation for delivery inspection. The preparation for delivery inspection shall be performed on all components of the sling system which have passed inspections specified herein and have been prepared for delivery. Inspection shall ensure compliance with the requirements of Section 5 of this specification.

5. PACKAGING

5.1 Preservation - packaging. The preservation - packaging shall be level A, B, or C as specified (see 6.2).

5.1.2 Level A. Each sling component shall be preserved and pack-aged in accordance with Method III of MIL-P-116, in a wood-cleated, plywood box conforming to overseas type of PPP-B-601.

5.1.3 Level E. Each sling component shall be preserved and pack-aged as for level A except that it shall be placed in a fiberboard container conforming to V13C of PPP-B-636.

5.1.4 Level C. Each sling component shall be preserved and pack-aged in accordance with ASTM D3951-82. The unit container shall be adequate as a shipping container.

5.2 Packing. Packing is not required.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked for shipment in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The external cargo helicopter sling systems are The external cargo helicopter sling systems are intended for use in transporting attachable cargo beneath suitably

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equipped helicopters. Weigh of transported cargo shall not exceed sling-load capacity.

*6.2 Ordering data.

*6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Level of preservation, packaging and packing (see 5.1).

*6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a Contract Data Requirements List (CDRL) (DD Form 1423), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Para No.	Data Requirements Title	Applicable DD No.	Option
4.1	Quality Assurance Program Plan-Requirements	DI-R-4802	
4.3	First Article Inspection Procedure	DI-T-4901	
4.3	First Article Inspection Report	DI-T-4902	
4.4	Production/Acceptance Inspection, Procedure	DI-T-4903	
4.4	Production Inspection Reports	DI-T-4904	
4.7.2	Test Procedures	DI-T-5248	

(Data item descriptions related to this specification and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, ASMDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 First article. Any changes or deviations of production sling assemblies from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the supplier of his obligation to furnish sling assemblies conforming to this specification.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-29398C(MC)	40,000 Pound Capacity Sling, System, Helicopter: External Cargo

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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